



LIFERISE INITIATIVE

(A Blockchain-Based Economic Empowerment Ecosystem for Stroke Survivors in Zambia)

BY

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Collegiate of Science in Blockchain Studies

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Capstone Date: 29 July 2026

ZAMBIA

A CAPSTONE PROJECT

SUBMITTED TO THE FACULTY OF BLOCKCHAIN STUDIES AND ARTIFICIAL
INTELLIGENCE AT THE ALTHASH UNIVERSITY
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR
THE COLLEGIATE OF SCIENCE IN BLOCKCHAIN STUDIES

CHICAGO, ILLINOIS.

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Authentication and Authorization
Data Management and Database Responsibilities
Blockchain Layer and Transaction Lifecycle
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3. Abstract

LifeRise Initiative is a proposed digital ecosystem intended to support the economic empowerment of stroke survivors in Zambia. The project uses blockchain selectively, rather than treating blockchain as a replacement for a normal database or backend. The architecture is hybrid: a mobile application provides the user interface; the backend controls application workflows and authorization; PostgreSQL stores private and frequently changing information; and an EVM-compatible blockchain provides a public, tamper-resistant record for selected transactions and utility-token activity.

The most important engineering principle is separation of concerns. Personal information remains off-chain. Blockchain is used only where an independently verifiable record, deterministic execution or token accounting creates a meaningful benefit. The LifeRise Token (LRT) is a utility token intended to support ecosystem participation and rewards. It is not designed to be digital money, a stablecoin, or a 1:1 representation of the Zambian Kwacha, US Dollar or another currency.

This documentation expands on the presentation slides and records the reasoning, technical workflow, security assumptions, data boundaries and blockchain design decisions behind the LifeRise platform.

4. Capstone Defense Approval Form

The Capstone Defense Approval Form is obtained from the Learning Management System (LMS). It is completed and signed by the appropriate academic authorities prior to the defense and forms part of the official submission package.

CAPSTONE DEFENSE APPROVAL FORM

Name:

Capstone Defense Approval Date & Time:

Degree: Nanodegree **Collegiate of Science**

Unit: College of Continuing and Professional Studies (CAPS)

Capstone Title:

Abstract:

In partial fulfillment of the requirements for the nanodegree of

(program)

We, the undersigned, recommend that the capstone project completed by the student listed above be acknowledged and counted as a requirement for graduation.

COMMITTEE APPROVAL

adviser name

signature

member

signature

member

signature

*At least one of the signatures above must be that of a member of the Althash University Faculty. By completing this box, you are confirming that the student above has satisfactorily completed the academic work for the program stated above. The submission of this form indicates the approval of the format and content of this document. This form is required for completion of the capstone project deposit.

5. CODE 499 Waiver Request Form

The CODE 499 Waiver Request Form is obtained from the LMS. It documents any approved fee waiver related to the capstone requirements and forms part of the official submission package.



CODE 499 WAIVER REQUEST FORM	
Name:	
Capstone Defense Approval Date & Time:	
Degree: Nanodegree	
Unit: College of Continuing and Professional Studies (CAPS)	
Capstone Title:	
Abstract:	
<i>I, the undersigned, request that the student indicated above be granted a fee waiver for the capstone project he or she completed on time and in accordance with cohort standards.</i>	
APPROVAL	

adviser name

signature

6. Introduction

The LifeRise Initiative addresses a social-impact problem through a technology architecture that combines established software engineering practices with blockchain concepts studied during the Blockchain Studies program.

Blockchain is often presented as a solution to every data problem. LifeRise instead treats blockchain as one component in a larger system. This distinction demonstrates practical understanding: a blockchain has strengths—integrity, shared verification and deterministic contract execution—but it also has limitations such as public visibility, gas costs, latency, and inability to directly query ordinary databases.

Project Objective

The objective is to design a platform that can support transparent participation, selected rewards and accountable transactions while protecting private user information and maintaining a realistic technical architecture.

7. Problem Statement

Stroke survivors can face reduced employment opportunities, financial dependency, limited access to support and difficulty participating in sustainable economic activities. Support programs may also need better mechanisms for tracking participation, documenting transactions and demonstrating accountability.

The problem is therefore not simply a lack of technology. It is the need for a trustworthy system that can combine privacy-sensitive application data with transparent and verifiable records where transparency is useful.

8. Methodology & Solution to the Problem

LifeRise proposes a platform through which beneficiaries and other ecosystem participants can register, participate in approved programs, receive support or rewards, and interact with participating organizations.

The solution uses three complementary mechanisms:

- **Conventional application infrastructure** for accounts, user interfaces, business workflows and private data.
- **Blockchain infrastructure** for selected verifiable transactions and token state.
- **Smart contracts** for deterministic rules that genuinely benefit from on-chain execution.

The design intentionally avoids putting medical or personally identifying information on a public blockchain. The architecture is hybrid because a fully on-chain design would expose sensitive information and increase costs, while a fully off-chain design would lose the transparency and independent verification benefits that blockchain can provide.

9. Vision, Mission, Goals, and Objectives

Vision

A Zambia in which stroke survivors regain economic independence, participate meaningfully in community life, and access transparent, accountable support systems enabled by appropriate technology.

Mission

To design and deliver a hybrid blockchain-enabled platform that empowers stroke survivors through verified participation, programmable incentives, privacy-preserving data management, and accountable transaction records.

Goals

- Enable secure registration and role-based participation for beneficiaries, volunteers, donors and administrators.
- Provide transparent, auditable reward and participation mechanisms via a utility token (LRT).
- Keep all personal and medical information off-chain while using blockchain only for verifiable state.
- Demonstrate a technically defensible, security-conscious architecture suitable for real-world pilot deployment.

Objectives

- Complete a working hybrid architecture (mobile UI, backend, PostgreSQL, EVM smart contracts).
- Implement authentication, authorization and wallet-based signing without exposing private keys.
- Define and document token utility, supply policy and allocation categories prior to deployment.
- Conduct phased testing and a limited pilot before any wider public launch.

10. Token Name

LifeRise Utility Token (also referred to as the LifeRise Token).

Classification: **Utility Token**. It is intended to have utility inside the LifeRise ecosystem and to create a programmable incentive and participation mechanism rather than a parallel monetary system.

11. Token Ticker

LRT

The ticker LRT is used consistently across documentation, smart-contract interfaces and application displays.

12. Token Maximum Supply

12.1 What Maximum Supply Means

Maximum supply is the hard upper limit on the total number of LifeRise Utility Tokens (LRT) that can ever exist. Once that limit is reached, no additional tokens can be created. In a properly designed smart contract, this limit is enforced by code: any mint operation that would push total supply above the cap is rejected. The rule is deterministic and publicly verifiable on the blockchain.

Maximum supply is a design decision, not a marketing number. It is not chosen because a large figure sounds impressive. It is derived from expected participation, reward frequency, program duration, and long-term sustainability of the incentive model. The goal is predictability and controlled issuance, not

artificial scarcity for speculative purposes.

12.2 Why a Maximum Supply Is Useful

A defined cap prevents uncontrolled minting. Without a hard limit, a privileged key or a flawed mint function could create unlimited tokens, destroying the integrity of the reward system and any trust participants place in the token's accounting. A fixed maximum supply makes issuance rules transparent: observers can see both the current total supply and the remaining room under the cap.

For LifeRise, the token is a utility mechanism for participation and rewards, not a currency and not a stablecoin. The maximum supply therefore supports a clear, auditable incentive budget. Participants and partners can understand that rewards are drawn from a finite, rules-based pool rather than from an open-ended printing process.

12.3 How the Cap Is Enforced

Enforcement belongs inside the smart contract. The contract maintains a total-supply counter. Every mint function checks that (current total supply + amount to mint) does not exceed the configured maximum. If the check fails, the transaction reverts. No off-chain process, administrator, or backend can override this rule once the contract is deployed with a fixed cap.

If the design instead uses a fixed initial supply with no further minting after deployment, that policy is stated clearly in the deployment documentation and the contract simply has no public mint function (or the mint function is permanently disabled). In either case—hard cap with controlled minting, or fixed supply with no post-deployment minting—the outcome is the same: total supply cannot grow beyond the published limit.

12.4 How the Number Should Be Chosen

The numerical value of the maximum supply should be justified by a simple model, not invented. The model considers: expected number of active beneficiaries and volunteers over the pilot and early scaling phases; average reward size per verified activity; frequency of rewardable activities; planned program duration; and a reserve for partnerships, community incentives, and operational contingencies. The resulting figure is then rounded to a clean, communicable number and locked into the contract.

Because the pilot scale and exact reward schedule are still being finalized, the precise numerical cap remains a tokenomics parameter under final design. Presenting an arbitrary large number as if it were already approved would be less honest and less defensible than stating the method by which the number will be derived and enforced.

12.5 Allocation Categories (Once the Cap Is Set)

Before deployment, the total supply is allocated across defined categories. Typical categories for LifeRise include:

- **Ecosystem rewards** — tokens reserved for verified participation by beneficiaries and program activities.
- **Community and volunteer incentives** — recognition for volunteers and community contributors under program rules.
- **Partnerships** — allocations that support collaboration with organizations that deliver or fund services.
- **Development and operational reserves** — a limited reserve for platform development, testing, and contingency, subject to clear governance.

Each category has a defined purpose. Where appropriate, release or vesting rules are documented so that large allocations cannot be dumped into circulation without process. Allocation does not change the maximum supply; it only decides how the fixed total is distributed over time.

12.6 What Maximum Supply Is Not

Maximum supply does not mean the token has a fixed monetary value. LRT is not pegged to the Zambian Kwacha, the US Dollar, or any other currency. The cap does not turn LRT into money, a stablecoin, or a

guaranteed store of value. It is a supply constraint on a utility token used for participation and rewards inside the LifeRise ecosystem.

The platform's core support functions are designed to remain operable even if the token component is later adjusted or de-emphasized. The token is an optional incentive layer, not the sole foundation of the service.

12.7 Defense Position (Summary)

If asked for the exact number today: the maximum supply is a tokenomics parameter under final design. It will be derived from expected participation, reward frequency, program duration, and sustainability, then locked into the smart contract as a hard, code-enforced cap (or as a fixed initial supply with no further minting). Presenting a method and an enforcement plan is more rigorous than inventing a number solely for the slides. Once the pilot parameters are approved, the numerical cap will be documented, allocated across the categories above, and deployed with the contract so that the limit is publicly verifiable and irreversible by ordinary administrative action.

13. Project Budget and Budget Allocation

For the pilot concept, the working budget is **K36,000** (Zambian Kwacha), corresponding to approximately **US\$1,800** at a working conversion rate of K20 per US\$1.

Category	Amount (ZMW)	Amount (USD)	Purpose
Survivor payments / support	K27,000	US\$1,350	Direct support and rewards for beneficiaries
Coordination	K6,000	US\$300	Project coordination and operational oversight
Communication	K3,000	US\$150	Outreach, materials and stakeholder communication
Total	K36,000	US\$1,800	Pilot working budget

Important: These figures should be replaced if the final approved budget differs. The final submission uses one consistent set of figures across slides and this PDF.

14. Token Slogan & Description

Slogan / Tagline

“Empowering Stroke Survivors. Restoring Independence. Creating Opportunity.”

Description

LRT is a utility token designed to reward verified participation in approved LifeRise activities, recognize contributions by volunteers or community participants, provide access to selected platform features or participation tiers, support transparent and auditable incentive distribution, and create a measurable participation record associated with an ecosystem wallet.

What LRT is NOT:

- It is not digital money for general everyday payments.
- It is not a stablecoin.
- It is not backed or pegged 1:1 to ZMW, USD or another fiat currency.
- It should not be described as a guaranteed monetary store of value.

15. Project Launch Date

The project follows a controlled, phased implementation rather than an immediate full public deployment.

- **Phase 1** – Requirements and research
- **Phase 2** – Architecture and design
- **Phase 3** – Development (mobile interface, backend, database, smart-contract prototype)
- **Phase 4** – Testing (authentication, authorization, contract functions, edge cases)
- **Phase 5** – Pilot (limited group, feedback, usability and operational measurement)
- **Phase 6** – Evaluation and scaling

Capstone presentation and documentation date: **29 July 2026**. A specific public pilot launch date will be set after successful completion of testing and internal evaluation.

16. Token Logo

The official LifeRise Initiative visual identity features a stylized figure rising from an open hand within a circular emblem, rendered in purple and green. The wordmark “LifeRise” appears with the subtitle “INITIATIVE ZAMBIA” and the tagline “Empowering Stroke Survivors. Restoring Independence. Creating Opportunity.”



High-resolution logo files are maintained in the project repository and branding package for consistent use across the mobile application, documentation and presentation materials.

17. GitHub Account & Link

GitHub provides version control, project organization and evidence of development. The repository contains a clear README describing the project, its purpose, architecture and current status. A logical structure separates documentation, smart contracts, backend services and mobile application code. Commit history shows meaningful development progress.

GitHub Repository:

<https://github.com/mitchello2022-png/LifeRise-initiative-.git>

For the capstone defense, the repository link and a screenshot of the repository home page provide evidence that the project artifacts are organized and accessible.

18. List of Courses Taken

The academic component of the capstone is supported by completed Blockchain Studies coursework as listed below.

BCE 501 – Blockchain Continuing Education™ (Series)

- BLKN 400 – The History of Blockchain Technology
- BLKN 205 – Blockchain Theory & Practice
- BLKN 300 – Blockchain Technology & Innovation
- CRPT 200 – Introduction to Cryptocurrency
- BLKN 215 – Applied Cryptography: Private & Public Keys and Digital Signatures
- BLKN 216 – Cryptography and Hash Functions
- CRPT 305 – Currencies, Tokens, and Stable Coins
- BLKN 320 – Consensus Mechanisms
- BLKN 334 – Wallet Safety and Security
- BLKN 232 – Interoperability
- BLKN 336 – Scalability and Other Challenges
- BLKN 420 – Decentralized Model and Consensus Mining
- BLKN 344 / DAPP 312 – Enterprise Blockchain
- BLKN 218 – Blockchain Anatomy, Nodes & Networks
- BLKN 194 – Independent Research in Blockchain Technology
- CRPT 312 – Tokenomics: Usage, Utility & Value
- BLKN/PROG 346 – Repository Systems
- BLKN/PROG 350 – Althash Blockchain
- BLKN/PROG 348 – Blockchain Architecture
- PROG 100 – Introduction to Smart Contracts
- BLKN/PROG 352 – Ethereum Blockchain
- PROG 358 – Introduction to Hyperledger Fabric
- TKNS 330 – NFT Development
- BLKN 356 – Self-Sovereign Identity
- BLKN 354 – Blockchain Leadership & Management
- BLKN 495 – Blockchain Profession
- BLKN 596 – Blockchain Career Development
- BLKN 492 – Directed Research in Blockchain Studies (Capstone Proposal)

-
- BLKN 499 – Capstone Defense

Additional Topics Covered:

- Blockchain-Enabled Electronic Health Records
- Cryptographic Hash Functions
- Private & Public Keys and Digital Signatures
- Blockchain Technology & Innovation Trends
- Diversity and Inclusion in Technology
- Blockchain Challenges: Scalability and Interoperability

B-Individual Requirements

19. Microcredentials & Link

Supporting microcredentials obtained during the Blockchain Studies program are included in the documentation package. Links and verification details are provided with the official credential documents.

[Attach microcredential certificates and verification links.]

20. Ladderized Certificates, Diploma & Link

The documentation package includes the Certificate of Satisfactory Competence in Blockchain Studies, Certificate of Outstanding Competence in Blockchain Studies, and the Collegiate of Science in Blockchain Studies Diploma.

Certificate of Satisfactory Competence – Awarded to Saviour Hanamwiinga on 17 May 2026 (48 Clock Hours, 80% Passing Score). Control Number: 6a0992f997e536d71d05dd68. In partial fulfillment of the requirements for the nanodegree of Blockchain Studies (CSC – BSTUD).



Certificate of Outstanding Competence – Awarded to Saviour Hanamwiinga on 27 Jun 2026 (96 Clock Hours, 80% Passing Score). Control Number: 6a3fc453fb3b4f28000545a7. In partial fulfillment of the requirements for the nanodegree of Blockchain Studies (CSC – BSTUD).



[Collegiate of Science in Blockchain Studies Diploma to be attached when available.]

21. Transcripts of Academic Records

Official Transcript of Records issued by Althash University for Saviour Hanamwiinga (Control Number: 6a4c1654f4cc4443280edf18; Date Issued: 6 Jul 2026; Total Semester Credit Hours: 15.9; Total Clock Hours: 159).

The transcript records Certificate of Satisfactory Competence and Certificate of Outstanding Competence in Blockchain Studies (80% passing).

Official Transcript – Page 1 of 3

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UNIVERSITY

OFFICIAL TRANSCRIPT OF RECORDS

NAME OF STUDENT: Saviour Hanamwiinga

DATE ISSUED: 6 Jul 2026

CONTROL NUMBER: 6a4c1654f4cc4443280edf18

TOTAL SEMESTER CREDIT HOURS 15.9

NANODEGREE EARNED:

*****NONE*****

TOTAL CLOCK HOURS: 159

CODE	TITLE	CLOCK HOURS	MASTERY
BLKN 400	THE HISTORY OF BLOCKCHAIN TECHNOLOGY	6.0	PASSED
BLKN 194	INDEPENDENT RESEARCH IN BLOCKCHAIN TECHNOLOGY	4.5	PASSED
BLKN 205	BLOCKCHAIN THEORY & PRACTICE	4.5	PASSED
BLKN 300	BLOCKCHAIN TECHNOLOGY & INNOVATION	4.5	PASSED
CRPT 200	INTRODUCTION TO CRYPTOCURRENCY	4.5	PASSED
BLKN 215	APPLIED CRYPTOGRAPHY: PRIVATE & PUBLIC KEYS AND DIGITAL SIGNATURE	4.5	PASSED
BLKN 216	CRYPTOGRAPHY AND HASH FUNCTIONS	4.5	PASSED
CRPT 305	CURRENCIES, TOKENS, AND STABLE COINS	4.5	PASSED
BLKN 320	CONSENSUS MECHANISMS	4.5	PASSED
BLKN 340	DIVERSITY AND INCLUSION IN THE TECHNOLOGY INDUSTRY	4.5	PASSED
BLKN 334	WALLET SAFETY AND SECURITY	4.5	PASSED
BLKN 232	INTEROPERABILITY	4.5	PASSED
BLKN 336	SCALABILITY AND OTHER CHALLENGES	4.5	PASSED
BLKN 420	DECENTRALIZED MODEL AND CONSENSUS MINING	4.5	PASSED
BLKN 344/ DAPP 312	ENTERPRISE BLOCKCHAIN	4.5	PASSED
BLKN 218	BLOCKCHAIN ANATOMY, NODES, & NETWORKS	4.5	PASSED

*****EARNED: CERTIFICATE OF SATISFACTORY COMPETENCE IN BLOCKCHAIN STUDIES*****
73.5 TOTAL CLOCK HOURS; 80% PASSING; 7.35 TOTAL SEMESTER CREDIT HOURS.

===== END OF COURSE INFORMATION =====

 Althash University

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 <https://t.me/althashedu>

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1

ALTHASH UNIVERSITY

OFFICIAL TRANSCRIPT OF RECORDS

NAME OF STUDENT: Saviour Hanamwiinga

CONTROL NUMBER: 6a4c1654f4cc4443280edf18

NANODEGREE EARNED: ***** NONE*****

DATE ISSUED: 6 Jul 2026

TOTAL SEMESTER CREDIT HOURS: 15.9

TOTAL CLOCK HOURS: 159

CODE	TITLE	CLOCK HOURS	MASTERY
CRPT 312	TOKENAMICS: USAGE, UTILITY & VALUE	4.5	PASSED
BLKN/PROG 346	REPOSITORY SYSTEMS	4.5	PASSED
BLKN/PROG 350	ALTHASH BLOCKCHAIN	4.5	PASSED
BLKN/PROG 348	BLOCKCHAIN ARCHITECTURE	4.5	PASSED
PROG 100	INTRODUCTION TO SMART CONTRACT	4.5	PASSED
HEAL 308	SELF-CARE AND WELL-BEING IN THE DIGITAL AGE	4.5	PASSED
BLKN 480	ISSUES & TRENDS IN BLOCKCHAIN TECHNOLOGY	4.5	PASSED
BLKN 490B	SPECIAL TOPIC: CRYPTOGRAPHIC HASH FUNCTIONS	4.5	PASSED
BLKN 490C	SPECIAL TOPIC: PRIVATE & PUBLIC KEYS AND DIGITAL SIGNATURES	4.5	PASSED
BLKN 342	IMPOSTER SYNDROME IN BLOCKCHAIN TECHNOLOGY	4.5	PASSED
BLKN/PROG 352	ETHEREUM BLOCKCHAIN	4.5	PASSED
BLKN/COMD 310	TRIBALISM IN BLOCKCHAIN & CRYPTOCURRENCY	4.5	PASSED
BLKN 495	BLOCKCHAIN PROFESSION	4.5	PASSED
BLKN/DAPP 311	BLOCKCHAIN-ENABLED ELECTRONIC HEALTH RECORDS	4.5	PASSED
TKNS 330	NFT DEVELOPMENT	4.5	PASSED
BLKN 356	SELF-SOVEREIGN IDENTITY	4.5	PASSED
BLKN 354	BLOCKCHAIN LEADERSHIP & MANAGEMENT	4.5	PASSED
PROG 358	INTRODUCTION TO HYPERLEDGER FABRIC	4.5	PASSED

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2

ALTHASH
UNIVERSITY

OFFICIAL TRANSCRIPT OF RECORDS

=====

NAME OF STUDENT: Saviour Hanamwiinga

DATE ISSUED: 6 Jul 2026

CONTROL NUMBER: 6a4c1654f4cc4443280edf18

TOTAL SEMESTER CREDIT HOURS: 15.9

NANODEGREE EARNED:

***** NONE*****

TOTAL CLOCK HOURS: 159

=====

BLKN 596

BLOCKCHAIN CAREER DEVELOPMENT

4.5

PASSED

*****EARNED: CERTIFICATE OF OUTSTANDING COMPETENCE IN BLOCKCHAIN STUDIES*****

159 TOTAL CLOCK HOURS; 80% PASSING; 15.9 TOTAL SEMESTER CREDIT HOURS.

===== END OF COURSE INFORMATION =====

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<https://www.althashuniversity/>

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LifeRise Initiative – Capstone Documentation

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22. BCE 501 Blockchain Continuing Education™

BCE 501: Blockchain Continuing Education

Overview & Impact

The BCE 501 Series develops and demonstrates professional competence in blockchain technology by connecting foundational concepts with real-world applications through eight key assessment modules (501A–501H).

Key Core Domains

1. Blockchain Architecture & Cryptography

Understanding distributed ledger technology, network architectures, consensus mechanisms, cryptographic principles, and blockchain security practices.

2. Smart Contracts & Tokenomics

Designing automated blockchain protocols, developing smart contract concepts, understanding token supply models, utility design, and token identifiers (tickers).

3. Governance & Compliance

Exploring blockchain governance frameworks, data privacy principles, auditing practices, risk management, and regulatory alignment.

Key Takeaway

The BCE 501 Series demonstrates the ability to analyze, design, and manage secure, transparent, and auditable blockchain ecosystems. These competencies support the development of practical solutions such as **LifeRise Initiative: A Blockchain-Based Economic Empowerment Ecosystem for Stroke Survivors in Zambia**.

23. Resume & Link

RESUME

Michelo Saviour Hanamwinga

Computer Scientist | Blockchain Developer | Cybersecurity Professional

Founder – LifeRise Initiative

Founder & Executive Director – PulseNetra

Professional Profile

Computer Scientist and Blockchain Developer with academic training and practical experience in Blockchain Technology, Cybersecurity, Web3, Smart Contracts, and digital solutions. Passionate about applying emerging technologies to address real-world social and economic challenges, with a focus on transparency, privacy, and inclusive innovation.

Education & Professional Development

- **Collegiate of Science in Blockchain Studies** – Althash University
- **Nanodegree in Blockchain Studies (CSC – BSTUD)** – Althash University
 - Certificate of Outstanding Competence (96 Clock Hours, 80% Passing Score)
 - Certificate of Satisfactory Competence (48 Clock Hours, 80% Passing Score)
- **Diploma in Computer Science & Information Processing** – Evelyn Home College
- **Associate Degree in Computer Science**
- **Cybersecurity Fundamentals Certificate**
- **IBM Cybersecurity Studies**
- **Certificate in Poultry Management**

Extensive coursework in Blockchain Technology, Cryptography, Tokenomics, Smart Contracts, Ethereum, Hyperledger Fabric, Blockchain Architecture, Wallet Security, Consensus Mechanisms, and Decentralized Systems.

Technical Skills

Blockchain Development & Architecture • Smart Contracts • Web3 Technologies • Cybersecurity • Cryptography • Tokenomics & Utility Tokens • Database & Data Management • IT Support • Data Analysis • Project Management • Digital & Emerging Technologies

Leadership & Projects

- **Founder – LifeRise Initiative:** Blockchain-based economic empowerment ecosystem for stroke survivors in Zambia
- **Founder & Executive Director – PulseNetra:** Technology and community-development projects
- Research and development in Blockchain and Web3 applications

Career Objective

To leverage Blockchain, Cybersecurity, and Computer Science expertise to develop secure, innovative, and inclusive digital solutions that create meaningful social and economic impact.

24. Blockchain Oath of Profession

The full Blockchain Oath of Profession appears at the end of this document.

25. Capstone Defense Awareness Plan

The Capstone Defense Awareness Plan outlines preparation steps, key talking points, anticipated questions and demonstration readiness for the formal defense. It ensures the candidate can clearly articulate architecture decisions, security rationale, token design and the hybrid on-chain / off-chain boundary.

Platform and Technical Overview

The platform is understood as a set of cooperating layers. Each layer has a defined responsibility and a defined boundary. A secure system should not depend on one technology doing everything.

Layer	Primary responsibility	Should NOT be responsible for
Mobile app	User interaction, display, request initiation	Final authorization or secret-key custody
Backend	Business logic, API, authorization, database access	Authoritative blockchain state without verification
PostgreSQL	Private application data and workflow records	Public immutable transaction proof
Blockchain	Verifiable on-chain state and transaction history	Private profiles or medical records
Smart contract	Token accounting and predefined on-chain rules	Direct PostgreSQL queries, private data, full app logic

Detailed End-to-End System Workflow

Step 1 – Registration

User creates an application account. Backend validates data and creates an internal user record. Passwords are hashed with a strong algorithm; never stored as plain text.

Step 2 – Authentication

User proves knowledge of credential. Server verifies password hash and establishes an authenticated session or token. MFA can be applied for higher-risk accounts.

Step 3 – Application request

Authenticated user performs an action (e.g., enrolling or requesting a reward). Mobile app sends the request to the backend over HTTPS/TLS.

Step 4 – Backend validation

Backend checks permission, request validity, program existence and whether the action has already been processed. Client-side validation alone is not a security boundary.

Step 5 – Database lookup

Backend retrieves required records from PostgreSQL (internal identifier, role, program participation, reward eligibility). Sensitive information stays inside the controlled environment.

Step 6 – Blockchain decision

If the action requires a verifiable on-chain record or token operation, the backend prepares the blockchain interaction. Otherwise the action remains entirely off-chain.

Step 7 – Wallet authorization

Where the user must authorize a blockchain action, the user's wallet signs the transaction. The application never asks the user to reveal the private key.

Step 8 – Smart-contract execution

The network executes the smart-contract function. The contract checks its own on-chain rules (access control, balances, limits, maximum-supply constraint).

Step 9 – Confirmation and event

After inclusion and required confirmations, the application associates the transaction hash and relevant contract event with the user's off-chain record.

Step 10 – User feedback

The application displays transaction status or reward. The user can independently inspect the public record while the private database record remains protected.

Important boundary: *The smart contract does not reach backward into PostgreSQL to look up a user's medical record. If off-chain information must affect an on-chain decision, the system needs an explicit trusted mechanism to bring an appropriate result on-chain.*

Authentication and Authorization

7.1 Application Authentication

Application authentication answers the question: “Who is this user?” The platform supports email or username credentials combined with a password. Passwords are never stored in plain text. Instead, they are processed with a strong, adaptive password-hashing algorithm (such as bcrypt, scrypt, or Argon2) that includes a unique salt per user. This design resists offline cracking even if the password database is compromised.

After successful verification of the password hash, the server establishes an authenticated session. Sessions use secure, HTTP-only cookies or short-lived access tokens delivered over HTTPS/TLS. Session tokens are rotated, expire after a defined period of inactivity, and can be revoked server-side. Multi-factor authentication (MFA) can be required for privileged roles (administrators, coordinators) or higher-risk actions, adding a second factor such as a time-based one-time password or a push approval.

7.2 Authorization (Role-Based Access Control)

Authorization answers the question: “What is this authenticated user allowed to do?” LifeRise uses role-based access control (RBAC). Typical roles include beneficiaries (stroke survivors enrolled in programs), volunteers, donors or supporting partners, and administrators. Each role is associated with a defined set of permitted actions.

Authorization is enforced exclusively on the server. Hiding a button in the mobile interface is not a security control. Every privileged API endpoint re-checks the user’s role and, where necessary, object-level ownership (for example, a beneficiary may only view their own records). This prevents both accidental misuse and deliberate attempts to access another user’s data by calling the API directly.

7.3 Wallet Authentication and Blockchain Authorization

A blockchain wallet is controlled by a cryptographic private key. When a user must authorize an on-chain action (for example, claiming a reward that mints or transfers LRT), the user’s wallet produces a digital signature over the transaction. The signature proves that the holder of the corresponding private key authorized that specific message. The platform never asks for, collects, or stores the user’s private key. Private-key custody remains entirely with the user or a secure wallet provider they choose.

Administrative or privileged contract functions (such as minting under controlled conditions or updating reward parameters) require on-chain access-control checks inside the smart contract itself. Knowing the contract address or the ABI is not sufficient to call those functions successfully.

7.4 Security Independent of Secrecy

The security model deliberately assumes that an attacker may know the full architecture: API routes, database technology, smart-contract address, and general workflow. Security therefore rests on cryptography, access control, input validation, TLS, secure secret management, and on-chain authorization—not on keeping the design secret. This approach is more robust under real-world conditions where source code, documentation, or network traffic may eventually become visible.

7.5 Major Authentication Attack Vectors and Controls

- **Credential stuffing and password reuse:** Mitigated by strong password policy, rate limiting, MFA, and breached-password detection where appropriate.
- **Session theft:** Mitigated by secure cookie attributes, short expiry, rotation, and mandatory TLS.
- **Phishing:** Users are instructed to sign only clear, understandable wallet messages and transactions; the application never requests the private key itself.
- **Private-key compromise:** The platform never holds user keys. Higher-value administrative keys require stronger custody controls (for example, multi-signature or hardware-backed wallets).

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- **Broken access control:** Every privileged operation is checked server-side and, where relevant, again by the smart contract.

Data Management and Database Responsibilities

8.1 What Belongs in PostgreSQL (Off-Chain)

PostgreSQL stores information that is private, relational, frequently updated, or unnecessary for public verification. Examples include user accounts and credentials (hashed), contact details, roles and permissions, program enrolment records, internal identifiers, support and case notes, application workflow state, notifications, and reporting data. PostgreSQL provides relational integrity, indexing, transactional consistency, backups, and controlled access—properties that ordinary application data requires and that a public blockchain is not designed to provide.

8.2 What Belongs On-Chain

Only the minimum state required for the decentralized function is stored on-chain. This includes wallet addresses associated with ecosystem participation, token balances, reward or distribution transactions, timestamps, and events emitted by the smart contract. On-chain data is public and independently verifiable; therefore nothing that would identify a medical condition, contact details, or other sensitive personal attributes is placed there.

8.3 Linking the Two Worlds

The backend maintains an internal mapping between an application-level record and a blockchain transaction hash (and, where useful, the corresponding contract event). This allows the application to state, for example, that “reward record X corresponds to transaction Y” without ever publishing the underlying private profile. The mapping itself remains inside the controlled application environment.

8.4 Safe Handling Practices

- Use least-privilege database accounts so application components only receive the permissions they need.
- Use parameterized queries or a trusted object-relational mapper (ORM) to eliminate SQL-injection risk.
- Encrypt sensitive fields at rest where appropriate and always protect data in transit with TLS.
- Maintain regular backups and test restoration procedures.
- Log security-relevant actions without logging passwords, private keys, or other secrets.
- Restrict administrative database access and apply data-minimization and retention rules.

Blockchain Layer and Transaction Lifecycle

A blockchain transaction is fundamentally different from a database write. A state-changing transaction is created, signed with the relevant private key, broadcast to the network, validated by the network's consensus mechanism, included in a block, and then confirmed according to the network's finality rules.

Typical transaction sequence

1. A user or authorized service creates the transaction call (function selector and arguments).
2. The transaction is signed with the relevant private key.
3. The signed transaction is broadcast to blockchain nodes.
4. The network validates the transaction and its signature.
5. The transaction is included in a block according to the network's consensus process.
6. The smart contract executes deterministically against the current blockchain state.
7. The application observes the transaction receipt and any emitted contract events.

A transaction can fail even though it was correctly signed. Common failure reasons include unmet require conditions inside the contract, the caller lacking permission, insufficient token balance, or the transaction running out of gas. The backend therefore treats "transaction submitted," "transaction included," and "transaction successful" as distinct states and handles each explicitly. Users receive clear feedback when an action is pending, confirmed, or rejected.

Smart Contract Design, Necessity and Limitations

10.1 Why Use a Smart Contract?

A smart contract is justified when LifeRise needs a shared, independently verifiable rule set that multiple parties can inspect and that executes deterministically without relying on a single trusted operator. Token accounting and predefined reward rules are clear examples: the rules can be audited, the supply constraints are enforced by code, and the resulting balances are publicly observable.

10.2 What the Contract Should Do

- Maintain token balances and permit only the token operations defined by the design.
- Apply predefined reward rules that are suitable for on-chain execution.
- Enforce access control on every privileged function.
- Emit events that applications and independent observers can consume.
- Enforce a maximum token supply if the tokenomics model adopts a hard cap.

10.3 What the Contract Should Not Do

- Store medical records, personal profiles, or any other sensitive private data.
- Query PostgreSQL or any external database directly.
- Replace the entire backend or attempt to implement full application logic on-chain.
- Assume that nobody will call its public functions; public functions are callable by anyone unless access control is enforced.
- Perform unnecessary on-chain calculations or storage writes that increase gas cost without adding verifiable value.

10.4 Gas and Efficiency

Every state-changing operation consumes network resources and therefore incurs gas. This creates a direct engineering trade-off. If a calculation can be performed safely off-chain and only its verifiable result needs to reach the blockchain, writing every intermediate step on-chain is inefficient and expensive.

LifeRise therefore minimizes on-chain work to the operations that genuinely benefit from public, deterministic execution.

10.5 Contract Security

The contract is deliberately kept small and auditable. Privileged functions use explicit access-control modifiers. State transitions are checked carefully. Before any production deployment the contract is tested against unauthorized calls, repeated calls, invalid inputs, arithmetic edge cases, and other relevant vulnerability classes. External review or formal audit is recommended once the final design is frozen.

Security Model and Attack-Vector Analysis

The security model assumes that attackers may know the architecture. Security therefore comes from cryptographic protections, access controls, secure implementation, and operational controls—not from hiding the system design. The following table summarizes the principal attack vectors considered for LifeRise and the primary controls that address each one.

Attack vector	What an attacker tries	Primary control
Credential attack	Steal/reuse passwords	Strong hashing, MFA, rate limiting, secure sessions
SQL injection	Manipulate database queries	Parameterized queries / ORM, validation
Broken authorization	Access another user's records	Server-side RBAC and object-level checks
API abuse	Replay or automate requests	Rate limiting, validation, idempotency, monitoring
Private-key theft	Authorize blockchain actions	Never collect keys; secure wallet/key custody
Unauthorized contract call	Invoke privileged function	On-chain access-control checks
Duplicate reward	Claim same reward repeatedly	Unique activity IDs, state checks, nonces/idempotency
Contract bug	Exploit token/reward logic	Minimal code, tests, review/audit before deployment
Privacy leakage	Read sensitive public data	Keep sensitive data off-chain; minimize on-chain metadata

In addition to the controls above, the system applies defence-in-depth: input validation at every boundary, least-privilege principles for both database and contract roles, continuous monitoring of anomalous API behaviour, and clear operational procedures for key rotation and incident response. The hybrid architecture itself is a security decision—by keeping private data off-chain, the surface area of public exposure is substantially reduced.

Conclusion

LifeRise demonstrates a practical application of blockchain rather than a blockchain-for-everything approach. The project separates private application data from public, verifiable blockchain state.

The mobile application and backend remain responsible for normal application operations. PostgreSQL handles private structured data. Blockchain provides a tamper-resistant transaction layer, while smart contracts automate only the rules that genuinely benefit from on-chain execution.

LRT is positioned as a utility token that supports participation and rewards. It is deliberately not a stablecoin, not a 1:1 representation of fiat currency and not the sole foundation of the platform. The resulting architecture is intended to be secure, privacy-aware, economically conscious and technically defensible.

BLOCKCHAIN OATH OF PROFESSION

We, Blockchain Professionals, solemnly pledge to uphold the highest standards of professionalism and ethics in the realm of blockchain technology.

Recognizing the transformative potential of blockchain to support economic empowerment, transparency, and accountability—particularly for vulnerable communities such as stroke survivors—we commit to embracing its principles and practices responsibly.

We pledge to acquire and apply the knowledge and skills needed to design and deploy blockchain solutions that protect privacy, ensure data integrity, and deliver real social impact. We will stay informed about the latest developments in blockchain, continuously update our expertise, and adapt our practices to incorporate blockchain only where it creates meaningful benefit.

We will prioritize accuracy, transparency, and integrity in our use of blockchain technology, ensuring the confidentiality and security of sensitive personal and health-related information. We will keep private data off-chain, enforce strong access controls, and never expose users to unnecessary risk.

We will adhere to regulatory requirements and promote ethical behavior in the design and operation of systems such as the LifeRise Initiative. With this pledge, we dedicate ourselves to the responsible and ethical advancement of blockchain technology in service of human dignity, independence, and opportunity.

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